# David Cropper

## AI Engineer | Applied AI · Agentic Systems · Workflow Automation

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## PROFESSIONAL PROFILE

AI Engineer building practical AI applications and governed business workflows using Python, LangGraph, FastAPI, n8n and LLM APIs. Independent engineering projects span multi-agent orchestration, retrieval-augmented generation (RAG), support automation, financial validation and operational intelligence. Brings prior commercial enterprise Data Engineering, software testing and test automation experience, with a strong focus on reliable integration, deterministic controls, evaluation and human oversight.

## SELECTED ENGINEERING STRENGTHS

- **Agentic architecture:** specialist agents, supervisor-directed investigation, bounded orchestration, evidence-gated decisions and interrupt/resume approvals.
- **Workflow integration:** n8n orchestration across LLM APIs, Slack, Gmail, PostgreSQL, forms, webhooks and HTTP service endpoints.
- **Engineering controls:** structured outputs, SQL reconciliation, idempotency, duplicate detection, policy enforcement, failure handling and owner approval.
- **Cloud and delivery:** Python/FastAPI services, Docker, PyTest, Git and hands-on AWS migration using ECS/Fargate, ECR, S3, IAM and CloudWatch.

## TECHNICAL SKILLS

**AI / Agents:** OpenAI APIs, LangGraph, LangChain, LLM integration, tool invocation, RAG, embeddings, ChromaDB, structured outputs, evaluation, human-in-the-loop governance

**Automation / Integration:** n8n, Slack, Gmail, PostgreSQL, REST APIs, webhooks, forms, conditional routing, retries, Microsoft Copilot Studio and Power Automate (hands-on prototype)

**Software / Cloud:** Python, FastAPI, Pydantic, SQL, Pandas, Docker, Git, PyTest, AWS (S3, ECR, ECS/Fargate, IAM, CloudWatch), Jenkins, Apache NiFi, Apache Airflow

## INDEPENDENT AI ENGINEERING EXPERIENCE

### AI Engineer — Independent R&D | Chase Risk & Compliance

Dec 2025 – Present

Portfolio-led independent engineering · Melbourne

Designs and develops AI-enabled applications and operational workflows, separating probabilistic interpretation from deterministic execution, validation and approval. Produces testable solutions and public project demonstrations; work is independent engineering rather than a claim of long-term commercial AI employment.

## FEATURED PROJECT — GOVERNED AGENTIC SUPPORT ORCHESTRATION PLATFORM

Python · LangGraph · FastAPI · n8n · Slack · LLM APIs · mock enterprise systems

- Built an enterprise-style support workflow for a failed booking and duplicate-payment scenario, combining specialist evidence, booking/payment, policy and reconciliation agents with a supervisor that can dynamically request further investigation.

- Orchestrated bounded agent reasoning with LangGraph, exposing case initiation and workflow continuation through FastAPI while n8n coordinates Slack notifications, approval collection and final operational updates.
- Separated LLM-proposed resolutions from deterministic controls validating payment target, amount, policy fit and evidence sufficiency before a human approve/reject decision using interrupt/resume.
- Applied least-privilege evidence exposure, fail-closed behaviour, loop bounds and explicit approval authority; **refund execution is intentionally simulated/not enabled**.

## APPLIED AI WORKFLOW AUTOMATION — NORTHSTAR PROJECTS

n8n · OpenAI · PostgreSQL · SQL · Slack · Gmail · HTTP APIs · simulated business systems

Designed four complementary business-process prototypes to demonstrate where LLM interpretation adds value, where exact business rules must remain deterministic, and how integrations, exception handling and approval controls fit together in operational workflows.

### Northstar — Support Intake & Triage

Processes incoming customer-support email to reduce manual reading, urgency assessment and routing, while retaining manual review for ambiguous or invalid inputs.

- Integrated Gmail intake with OpenAI-based structured classification, summarisation and urgency assessment in n8n.
- Applied deterministic Switch/IF routing to appropriate Slack support channels; handled incomplete or ambiguous cases through manual-review paths.
- Added API retry behaviour, failure notifications and an error workflow; exercised positive, negative and adversarial input scenarios.

### Northstar — Invoice Intake & Validation

Automates the intake of PDF invoices while keeping financial checks, deduplication and external submission decisions under conventional software control.

- Used OpenAI to extract structured invoice fields from PDFs, followed by deterministic required-field and arithmetic validation.
- Queried PostgreSQL to detect previously processed invoices; persisted idempotency keys and processing states to prevent repeat submission.
- Submitted validated payloads to a **simulated/test accounting API** and recorded failure cases with Slack exception notifications; no live accounting-system integration claimed.

### Northstar Logistics — Daily Revenue Validation

Automates routine financial data-quality assurance over source and summarised revenue datasets without relying on an LLM for exact calculations.

- Implemented scheduled or manually initiated n8n checks backed by PostgreSQL SQL queries for completeness, nulls, duplicate records and customer coverage.
- Compared source and aggregated revenue values for reconciliation, marking passing datasets as ready for downstream processing.
- Issued Slack exceptions on failed checks, deliberately using **deterministic SQL rather than AI** for repeatability and auditability.

### Northstar — Travel Exception Approval

Demonstrates a controlled approval process where AI interprets an employee's justification but cannot independently authorise an exception.

- Accepted requests through an n8n form, validated structured inputs and calculated policy-relevant facts.
- Used OpenAI to summarise or interpret justifications, while deterministic policy rules established approval authority and routing.
- Integrated Slack review with execution pause/resume and human APPROVE/REJECT capture; recorded decisions and notified requesters.
- Controlled prototype using simulated policy inputs; manager escalation, persistent enterprise audit storage and enterprise authentication are not claimed as implemented.

## ADDITIONAL SELECTED AI ENGINEERING PROJECTS

### Career Intelligence Copilot (CIC)

Python · Pydantic · OpenAI APIs · PyTest · Git

Built a decision-support system for a real AI Engineering job search, helping assess opportunities, decide which applications to pursue and prepare evidence-grounded application materials while leaving decisions with the user.

- Modelled a structured career profile and opportunity evidence, producing explainable assessment and application strategies with provenance.
- Combined LLM interpretation and drafting with deterministic workflow states, duplicate detection, validation and explicit owner review.
- Used regression/golden scenarios, acceptance gates and iterative calibration to improve reliability; outputs still require human review.

### Governance-Aware Document Intelligence RAG

Python · FastAPI · Docker · LangChain · ChromaDB · OpenAI · PyTest

Designed document-question answering for organisational policies, procedures and other internal knowledge, emphasising grounded answers and visibility into unsupported responses.

- Implemented ingestion of text, PDF and DOCX documents, embedding generation, retrieval and cited response construction.
- Introduced telemetry, evaluation and grounding controls so uncertain or unsupported responses could be identified rather than presented as established facts.
- Separated API/service functions from the user-facing experience, with automated tests of core behaviour.

### Operational Intelligence Copilot

Python · FastAPI · OpenAI APIs · PyTest

Built an analytics-led assistant for exploring business operations and explaining trends, variances, anomalies and risks using evidence from underlying datasets.

- Combined dataset profiling, deterministic calculations and analysis workflows with controlled LLM narrative generation.
- Kept quantitative findings in conventional analytic logic, using AI to make validated results easier for decision-makers to interpret.
- Regression-tested analytical outputs and structured explanations for traceability.

### Payroll Diagnostics Engine & AWS Deployment

Python · Pandas · Docker · S3 · ECR · ECS/Fargate · IAM · CloudWatch · PyTest

Engineered a rule-based payroll assurance workflow that examines workforce/payroll records for anomalies, reconciles inputs and creates explainable findings and executive reporting.

- Created deterministic diagnostics and reproducible report-generation pipelines, with controls to distinguish verified findings from possible anomalies.
- Migrated containerised processing from local execution to AWS using ECR, ECS/Fargate and client-scoped S3 storage with IAM and CloudWatch visibility.
- Compared cloud execution with a frozen local baseline, exercising synthetic success/failure scenarios and client-isolation checks.

### Public Holiday Entitlements Application

Python · Streamlit · SQLite · REST APIs · geospatial processing

Built an Australian public-holiday entitlement tool that uses location and applicable rules to determine relevant holiday information for payroll and HR workflows.

- Combined geocoding, local-government-area mapping and public-holiday API data with deterministic entitlement logic.
- Supported individual and batch inputs, with downloadable compliance-oriented reports.

## COMMERCIAL ENGINEERING EXPERIENCE

### Data Engineer — nbn Australia

Mar 2020 – Oct 2023

- Developed and supported enterprise data pipelines and operational reporting with AWS, Python, SQL and Apache NiFi; used Apache Airflow for orchestration.
- Applied PostgreSQL, data validation, automated testing, Git/Jenkins CI/CD and Agile delivery in production engineering environments.

### Earlier Commercial Roles — Test Analyst / Test Automation Engineer

Worked across enterprise software delivery, automation frameworks, regression testing and production support at Bakers Delight, Console and AccessHQ.

## PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

### AWS Certified Developer – Associate · Databricks Certified Data Engineer Professional

Professional development (Oct 2023 – Nov 2025): structured Data Engineering and Applied AI study, including Python, LLM application development, LangChain and cloud/data platforms. General Assembly Data Science Immersive (2019).